

Permutations, Combinations, Binomial Theorem

Subtitle

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OUTLINE

1. Basic Counting Principles
2. Pigeonhole Principle
3. Permutations
4. Combinations
5. Binomial Theorem

The Product Rule

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.

In sets, If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements in each set, i.e.

$$| A_1 \times A_2 \times \dots \times A_M | = | A_1 | \cdot | A_2 | \cdot \dots \cdot | A_m |$$

Example

The chairs of an auditorium are to be labelled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labelled differently?

Sol The procedure of labelling a chair consists of two tasks, namely, assigning to the seat one of the 26 uppercase English letters, and then assigning to it one of the 100 possible integers. The product rule shows that there are $26 \cdot 100 = 2600$ different ways that a chair can be labelled. Therefore, the largest number of chairs that can be labelled differently is 2600.

Example:

The chairs of a Lecture hall are to be labelled with a letter followed by a positive integer not exceeding 100.

What is the largest number of chairs that can be labelled differently?

Solution

- 26 different ways to select a letter.
- 100 ways to select a positive integer not exceeding 100.
- Using the product rule, $26 \times 100 = 2600$ chairs can be labelled.

Example:

How many number plates are available if each number plate begins with a fixed letter (B) followed by 3 digits and finally ending with 3 letters? No sequence of numbers and letters are prohibited. So B000AAA is a legal number plate.

Solution

- Since B is fixed, 1 way to select B.
- for each of the 3 digits they are 10 ways to select each, giving $10 \times 10 \times 10$ different ways.
- For each of the 3 letters, they are 26 ways to select each, giving $26 \times 26 \times 26$ different ways.
- So total is: $1 \times 10 \times 10 \times 10 \times 26 \times 26 \times 26 = 17, 576, 000$ number plates.

Example

How many different bit strings of length seven are there?

Sol Each of the seven bits can be chosen in two ways, because each bit is either 0 or 1. Therefore, the product rule shows there are a total of $2^7 = 128$ different bit strings of length seven.

Example - Counting Functions

How many functions are there from a set with m elements to a set with n elements?

Sol A function corresponds to a choice of one of the n elements in the codomain for each of the m elements in the domain. Hence, by the product rule there are $n \times n \times \cdots n = n^m$ functions from a set with m elements to one with n elements. e.g., there are $5^3 = 125$ different functions from a set with three elements to a set with five elements.

The Sum Rule

If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

The sum rule can be phrased in terms of sets: If $A_1, A_2, \dots, A_m$ are pairwise disjoint finite sets, then the number of elements in the union of these sets is the sum of the numbers of elements in the sets, i.e.

$$|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|$$

Example

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from.

Sol The student can choose a project by selecting a project from the first list, the second list, or the third list. Because no project is on more than one list, by the sum rule there are $23 + 15 + 19 = 57$ ways to choose a project.

The Subtraction Rule, aka The Principle of Inclusion-Exclusion

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

In sets, to count elements in $A \cup B$: $|A \cup B| = |A| + |B| - |A \cap B|$

Example: How many bit strings of length 8 either start with a 1 or end with two bits 00?

Solution

A_1 - Number of bit strings of length 8 starting with 1 is 2^7 .

A_2 - Number of bit strings of length 8 ending with 00 is 2^6 .

A_5 - Common bit strings. Those that begin with 1 and end with 00 is 2^5

So the total number of bit strings that start with a 1 or end with 00: $A_1 + A_2 - A_5$.

Total: $2^7 + 2^6 - 2^5$.

$$\begin{array}{c} 1 \text{ ---} \\ \underbrace{\hspace{10em}} \\ 2^7 = 128 \text{ ways} \end{array}$$

$$\begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} 0 \ 0 \\ \underbrace{\hspace{10em}} \\ 2^6 = 64 \text{ ways} \end{array}$$

$$\begin{array}{c} 1 \text{ ---} \text{---} \text{---} \text{---} 0 \ 0 \\ \underbrace{\hspace{10em}} \\ 2^5 = 32 \text{ ways} \end{array}$$

The Division Rule

There are $\frac{n}{d}$ ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

Example

How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbour and the same right neighbour?

Sol We arbitrarily select a seat at the table and label it seat 1. We number the rest of the seats in numerical order, proceeding clockwise around the table. Note that there are four ways to select the person for seat 1, three ways to select the person for seat 2, two ways to select the person for seat 3, and one way to select the person for seat 4. Thus, there are $4! = 24$ ways to order the given four people for these seats. However, each of the four choices for seat 1 leads to the same arrangement, as we distinguish two arrangements only when one of the people has a different immediate left or immediate right neighbour. Because there are four ways to choose the person for seat 1, by the division rule there are $\frac{24}{4} = 6$ different seating arrangements of four people around the circular table.

Tree Diagrams

A tree consists of a root, a number of branches leaving the root, and possible additional branches leaving the endpoints of other branches. To use trees in counting, we use a branch to represent each possible choice. We represent the possible outcomes by the leaves, which are the endpoints of branches not having other branches starting at them.

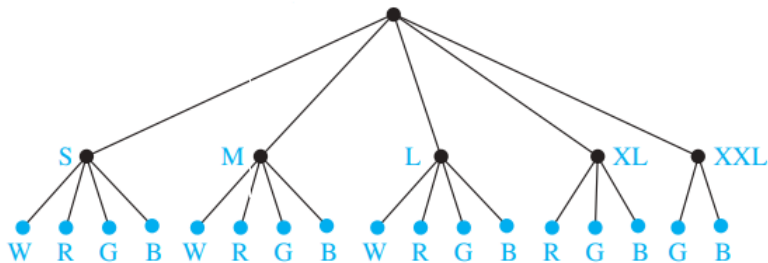
Example

Suppose that “UB” T-shirts come in five different sizes: S, M, L, XL, and XXL. Further suppose that each size comes in four colours, white, red, green, and black, except for XL, which comes only in red, green, and black, and XXL, which comes only in green and black. How many different shirts does a souvenir shop have to stock to have at least one of each available size and colour of the T-shirt?

Sol

The tree diagram below displays all possible size and colour pairs.

W = white, R = red, G = green, B = black



It follows that the souvenir shop needs to stock 17 different T-shirts.

The Pigeonhole Principle

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Example In any group of 27 English words, there must be at least two that begin with the same letter, because there are 26 letters in the English alphabet.

Example How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Sol There are 101 possible scores on the final. The pigeonhole principle shows that among any 102 students there must be at least 2 students with the same score.

The Generalised Pigeonhole Principle

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil \frac{N}{k} \rceil$ objects.

Example Among 100 people there are at least $\lceil \frac{100}{12} \rceil = 9$ who were born in the same month.

Permutations

- A **permutation** is a linear arrangement of objects where order or position matters. For any integer n with $n \geq 1$, the number of permutations of a set with n elements is $n!$.

Example 1: How many ways can the letters in the word COMPUTER be arranged in a row?

Example 2: How many ways can the letters in the word COMPUTER be arranged if the letters CO must remain next to each other (in order) as a unit?

Permutations of Selected Elements

Definition: An **r-permutation** of a set of n elements is an ordered selection of r elements taken from the set of n elements. The number of **r-permutations** of a set of n elements is denoted $P(n, r)$.

Theorem If n and r are integers and $1 \leq r \leq n$, then the number of r -permutations of a set of n elements is given by the formula

$$P(n, r) = n(n-1)(n-2)\dots(n-r+1)$$

or equivalently,

$$P(n, r) = \frac{n!}{(n-r)!}$$

Example 1: Evaluate $P(5, 2)$.

Example 2: How many 4-permutations are there of a set of 7 objects?

Example 3: Evaluate $P(5, 5)$

Example: How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other? *Hint: First position the men, then consider possible positions for the women.*

Sol: First consider the position of the men, then the position of the women (In general first consider the group with more people).

Possible ways to arrange the 8 men is $8! = 40320$.

Now, we need to find how many ways we can arrange 5 women in the 9 possible places, which is ${}^9P_5 = 9 * 8 * 7 * 6 * 5 = 15120$

Total number of possible arrangements satisfying both constraints is:

$40320 * 15120 = 609638400$.

Example: Suppose we have 13 chairs (in a row) with 9 students, and 4 lecturers to be seated. What is the number of seatings where every lecturer has a student to his/her immediate left and right?

Permutations involving Indistinguishable/Repeated Objects

If there are n objects, with n_1 of the first type, n_2 of the second type, ... , n_r of the r -th type, where $n_1 + n_2 + \dots + n_r = n$, then there are

$$\frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_r!}$$

distinguishable permutations/arrangements.

Example: Find the number of permutations of the letters of the word MISSISSIPPI.

Sol The word MISSISSIPPI has 11 letters. If the letters were all different there would have been $11!$ different permutations. But MISSISSIPPI has 4 S's, 4 I's, and 2 P's that are alike. So the answer is $\frac{11!}{4!4!2!} = 34,650$.

Example If a coin is tossed six times, how many different outcomes consisting of 4 heads and 2 tails are there?

Again, we have permutations with similar elements. We are looking for permutations for the letters HHHHTT. The answer is $\frac{6!}{4!2!}$

Counting Elements of Disjoint Sets

Example: Counting the Number of Integers between 100 and 999 that are Divisible by 7.

Combinations

The number of r -combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, and read as “ n choose r ” equals

$$C(n, r) = \frac{n!}{r!(n-r)!} = \binom{n}{r}$$

It denotes the number of subsets of size r (r -combinations) that can be chosen from a set of n elements.

Number of Bit Strings with Fixed Number of 1's

How many eight-bit strings have exactly three 1's?

Solution

To solve this problem, imagine eight empty positions into which the 0's and 1's of the bit string will be placed. In step 1, choose positions for the three 1's, and in step 2, put the 0's into place.

Once a subset of three positions has been chosen from the eight to contain 1's, then the remaining five positions must all contain 0's (since the string is to have exactly three 1's). It follows that the number of ways to construct an eight-bit string with exactly three 1's is the same as the number of subsets of three positions that can be chosen from the eight into which to place the 1's.

Thus, this equals

$$\binom{8}{3} = \frac{8!}{3!.5!} = \frac{8.7.6.5!}{3.2.5!} = 56$$

Example A university SRC consists of 15 students.

(i) In how many ways can a committee of six be selected from the membership of the council?

Sol:

The number of committees of 6 is

$$\binom{15}{6} = \frac{15!}{(15-6)!(6!)} = 5,005$$

(ii) Two council members have the same major and are not permitted to serve together on a committee. How many ways can a committee of six be selected from the membership of the council?

Sol:

Let the two SRC members with the same majors be A and B.

Then,

Number of committees that do not contain A and B together = [The number of committees with A and five others — none of them B] + [the number of committees with B and five others — none of them A] + [the number of committees with neither A nor B]

$$= \binom{13}{5} + \binom{13}{5} + \binom{13}{6} = 1287 + 1287 + 1716 = 4290$$

Alternatively, = [the total number of committees] - [number of committees that contain both A and B]

$$= \binom{13}{6} - \binom{13}{4} = 5005 - 715 = 4290$$

(iii) Two council members always insist on serving on committees together. If they cannot serve together, they won't serve at all. How many ways can a committee of six be selected from the council membership?

Sol:

[The number of committees with both A and B] + [The number of committees with neither A nor B]

$$\binom{13}{4} + \binom{13}{6} = 715 + 1716 = 2431$$

(iv) Suppose the council contains eight men and seven women,

(a) How many committees of six contain three men and three women?

Sol:

[The number of subsets of three men chosen from eight].[The number of subsets of three women chosen from seven]

$$\binom{8}{3} \cdot \binom{7}{3} = 1960$$

(b) How many committees of six contain at least one woman?

Sol:

[the number of committees with at least one woman] = [the total number of committees]
- [the number of all-male committees]

$$= \binom{15}{6} - \binom{8}{6} = 5005 - 28 = 4977$$

(v) Suppose the council consists of three 1st years, four 2nd years, three 3rd years, and five 4th years. How many committees of eight contain two representatives from each class?

Sol:

$$\binom{3}{2} \cdot \binom{4}{2} \cdot \binom{3}{2} \cdot \binom{5}{2} = 540$$

Proposition

For nonnegative integers $n \geq r$, we have .

$$\binom{n}{r} = \binom{n}{n-r}$$

Theorem

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Proof

There are $\frac{n!}{(n-r)!}$ r -permutations. For each r -combination, how many r -permutations does it correspond to? Since the r elements are fixed, they are just the r -permutations on r elements, which means the number is $r!$. So each r -combination appears in $r!$ r -permutations. And thus

$$\binom{n}{r} = \frac{n!}{(n-r)!} / r! = \frac{n!}{r!(n-r)!}$$

$$\binom{n}{1} = n, \binom{n}{n} = 1, \binom{n}{0} = 1$$

Example: One buys a Powerball lottery ticket as follows: select five distinct unordered numbers from 1 to 69 for the white balls, then select one number from 1 to 26 for the red Powerball. There is only one winning number of 5 white balls and 1 red ball for the top prize. How many possible outcomes of the winning numbers?

Solution:

To get a combination, there are two steps: choose five white balls and one red ball. So we apply the multiplication rule first. For the white balls, they are a 5 selection among 69 numbers, so the number of ways is $\binom{69}{5} \cdot 26 = 292,201,338$

Example: A club of 30 people wants to choose an executive board consisting of president, secretary, treasurer and two other officers. In how many ways can this be achieved?

Solution:

We break it into independent steps:

- . president, secretary and treasurer - $P(30, 3)$.
- . two other officers - $\binom{27}{2}$

So the answer is $P(30,3) \cdot \binom{27}{2} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26}{2}$

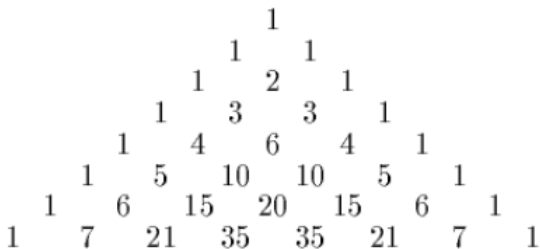
Pascals' formula

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$$

This is a recursive relation because all $\binom{n}{\cdot}$ values give all $\binom{n+1}{\cdot}$

The Pascal's triangle is a triangular array of numbers such that the n -th row are just the numbers.

$$\binom{n-1}{0}, \binom{n-1}{1}, \binom{n-1}{2}, \dots, \binom{n-1}{n-1}$$



Binomials

Definition: In algebra, a sum of two terms, such as $a + b$, is called a **binomial**.

Expand

$$(a + b)^4$$

The coefficients in the expansion of $(a + b)^4$ are the numbers "n choose r". $\binom{n}{r}$ are called **binomial coefficients**.

The Binomial Theorem

For any real numbers a , b and nonnegative integer n ,

$$\begin{aligned}(a + b)^n &= \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \\ &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n-1} a b^{n-1} + \binom{n}{n} b^n\end{aligned}$$

Each term in the expansion is of the form $a^{n-k} b^k$

Example: Expand the following expressions using the binomial theorem:

(a)

$$(a + b)^5$$

(b)

$$(x - 4y)^4$$

Example: Find the n -th term.

The general formula to find the n^{th} term of $(a + b)^n$ is

$$\binom{n}{r} a^{n-r} b^r$$

(a) Find the 7th term in

$$(3x^2 - 2y)^{13}$$

Example: Find the coefficient of the given term when the expression is expanded by the binomial theorem

(a) $a^5 b^7$ in $(a - 2b)^{12}$

(b) $p^{16} q^7$ in $(3p^2 - 2q)^{15}$

The End